

Fundamentals of IPv4 Addressing and Subnetting

Summary

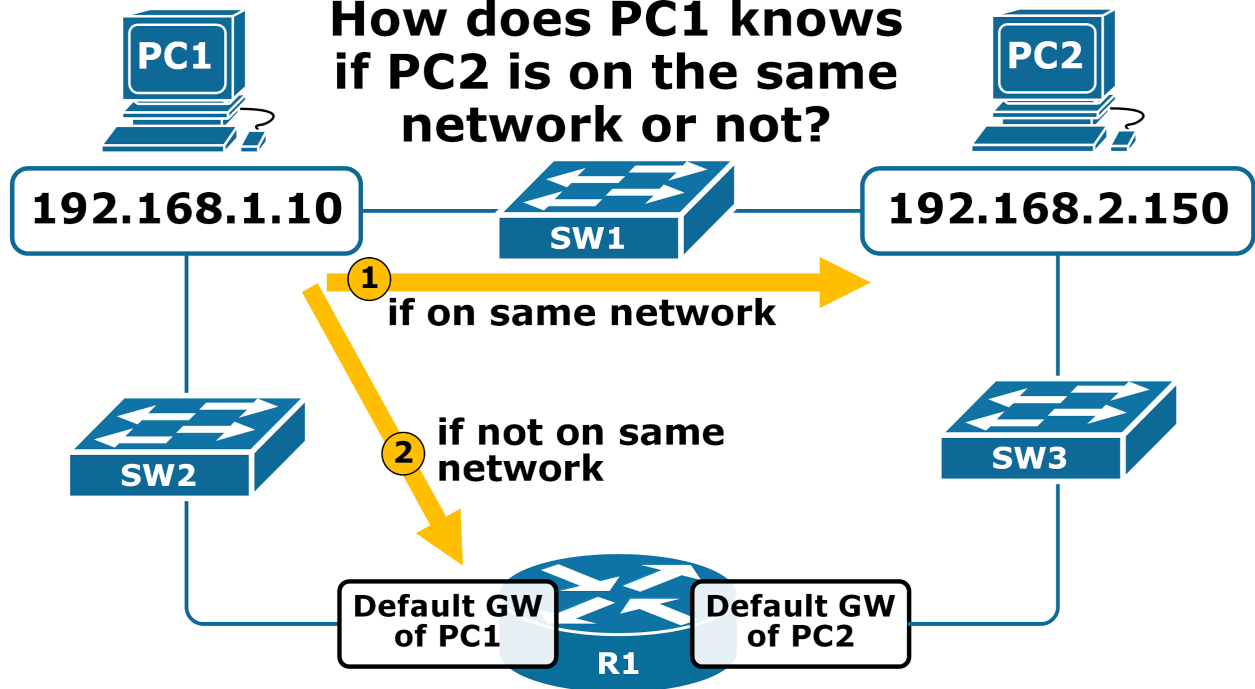
Subnetting is one of the most critical and exam-relevant topics in IPv4 networking, especially for **GATE and other competitive exams**, where it dominates addressing-related questions. Its purpose is to **efficiently manage IP address space**, reduce wastage caused by classful allocation, and allow logical division of networks based on organizational needs.

This briefing consolidates:

- How **network masks** separate Network ID and Host ID
- Identification of **special IP addresses** (Network ID, Direct Broadcast, Limited Broadcast)
- The **bit-borrowing mechanism** used to create subnets
- Practical techniques to compute **subnet ranges, host capacity, and broadcast addresses**

1. The Role and Structure of Network Masks

**Without a subnet mask:
How does PC1 know
if PC2 is on the same
network or not?**



IP Address:	192	.	168	.	100	.	1
IP (Binary):	11000000	.	10101000	.	01100100	.	00000001
	Network ID						Host ID
SM (Binary):	11111111	.	11111111	.	11111111	.	00000000
Subnet Mask:	255	.	255	.	255	.	0

A **network mask (subnet mask)** is a 32-bit binary value used to determine **which part of an IP address identifies the network** and **which part identifies the host**.

1.1 Meaning of Bits in a Mask

- Binary 1 (1) → Network ID (NID)

- **Binary 0 (0) → Host ID (HID)**

Example (Class C default mask):

IP Address : 192.168.1.10

Subnet Mask: 255.255.255.0

Binary Mask: 11111111.11111111.11111111.00000000

Here:

- First 24 bits → Network
- Last 8 bits → Host

1.2 Default (Natural) Masks by Class

Class	Network Bits	Host Bits	Default Mask
A	8	24	255.0.0.0
B	16	16	255.255.0.0
C	24	8	255.255.255.0

These are called **natural masks** because they are used *before subnetting*.

1.3 Finding Network ID (Bitwise AND)

To compute the **Network ID**:

Network ID = IP Address AND Subnet Mask

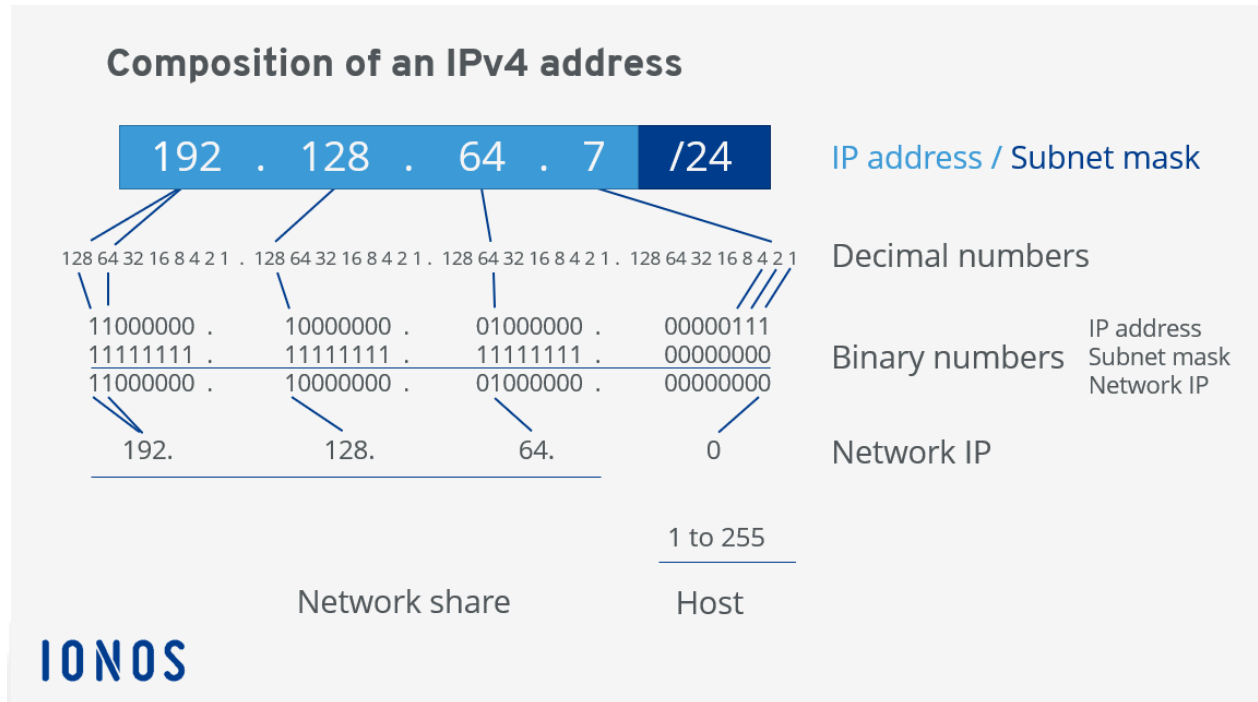
Key properties:

- Any value AND 255 = same value
- Any value AND 0 = 0

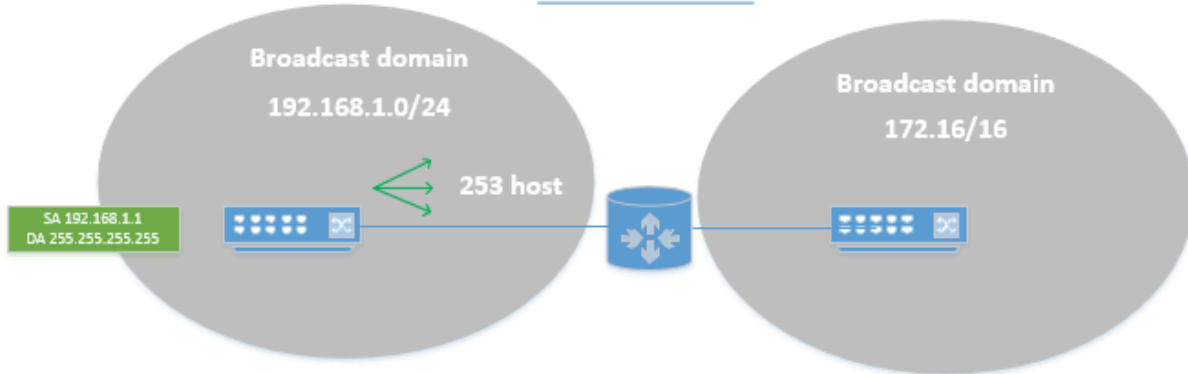
This is why:

- Mask octet = 255 → IP octet retained
- Mask octet = 0 → IP octet becomes 0

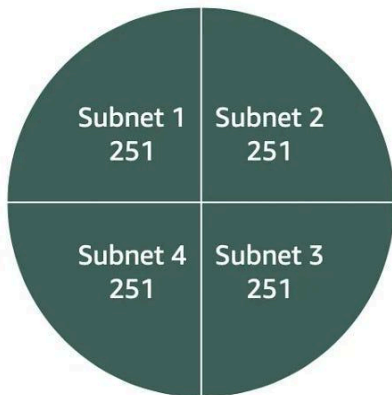
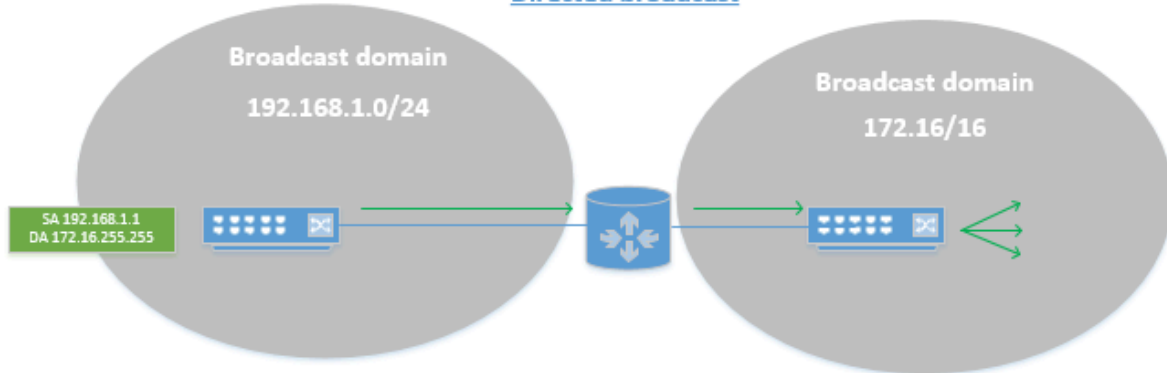
2. IP Address Classification and Special Addresses



Limited broadcast



Directed broadcast



IP address	Reserved for
10.0.0.0	Network address
10.0.0.1	VPC local router
10.0.0.2	DNS server
10.0.0.3	Future use
10.0.3.255	Network broadcast address



Reserved Subnet Addresses

all 0s – "this subnet"

all 1s – broadcast

Ex: 10.10.1.32/27

10.10.1.32

00001010.00001010.00000001.001xxxxx

27 bits

all 0s = 10.10.1.32 =

00100000

all 1s = 10.10.1.63 =

00111111

Not every IP address can be assigned to a host.

2.1 Network ID (NID)

- Host bits = all 0
- Identifies the **entire network**
- Cannot be assigned to a device

Example (Class C):

192.168.1.0 → Network ID

2.2 Direct Broadcast Address (DBA)

- Host bits = all 1
- Sends data to **all hosts in a specific network**
- Cannot be a source IP

Example (Class C):

192.168.1.255 → Direct Broadcast

2.3 Limited Broadcast Address (LBA)

- All 32 bits = 1

255.255.255.255

- Broadcasts to **all hosts in the local network**
- Independent of network class
- Destination-only address

2.4 Class D and Class E

- **Class D (224–239)** → Multicast
- **Class E (240–255)** → Experimental

These classes:

- Do **not** have Network ID / Host ID
- Cannot be subnetted
- Cannot be assigned to normal hosts

3. Why Subnetting Is Necessary

3.1 The Core Problem: Address Wastage

Classful allocation is rigid.

Example:

- Organization needs 2^{22} IPs
- Closest class = **Class A** (2^{24} IPs)
- **Wastage** = $2^{24} - 2^{22} = 12.5$ million IPs

👉 Subnetting solves this by dividing large blocks into smaller logical networks.

3.2 Borrowing Bits (Heart of Subnetting)

Subnetting works by:

- **Borrowing bits from Host ID**
- Using them as **Subnet ID**

If:

- n bits borrowed
- **Number of subnets = 2^n**

Borrowed Bits	Subnets
1	2
2	4
3	8
4	16

3.3 Impact on Host Capacity

If:

- Remaining host bits = h

Then:

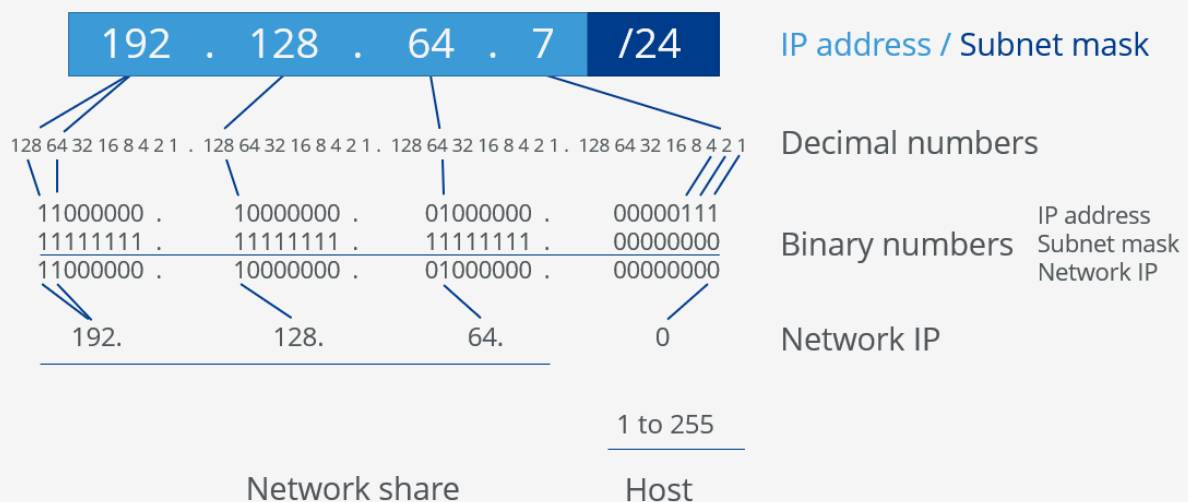
- **Total IPs per subnet = 2^h**
- **Valid hosts per subnet = $2^h - 2$**

(-2 for Subnet ID & Subnet Broadcast)

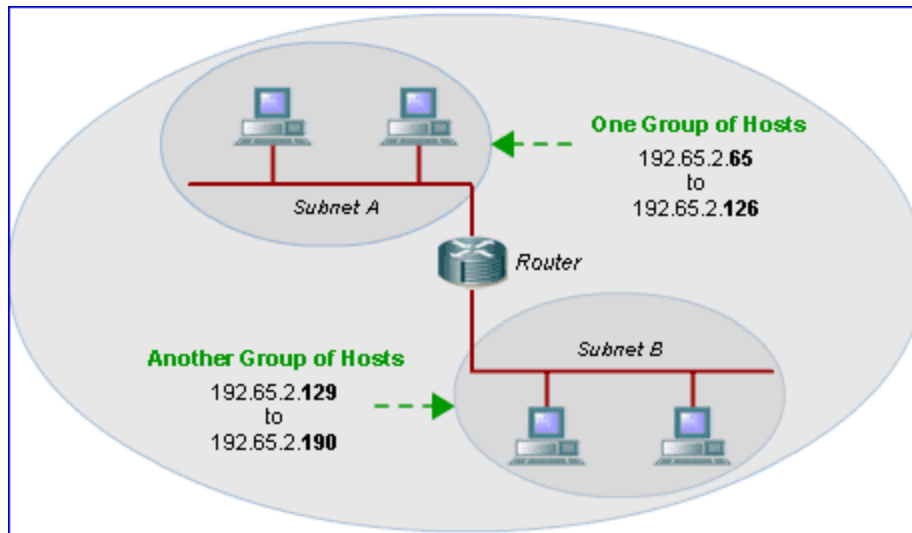
4. Subnet Range Determination (Practical Analysis)

Network Bits	Subnet Mask	Bits Borrowed	Subnets	Hosts/Subnet
16	255.255.0.0	0	0	65534
17	255.255.128.0	1	2	32766
18	255.255.192.0	2	4	16382
19	255.255.224.0	3	8	8190
20	255.255.240.0	4	16	4094
21	255.255.248.0	5	32	2046
22	255.255.252.0	6	64	1022
23	255.255.254.0	7	128	510
24	255.255.255.0	8	256	254
25	255.255.255.128	9	512	126
26	255.255.255.192	10	1024	62
27	255.255.255.224	11	2048	30
28	255.255.255.240	12	4096	14
29	255.255.255.248	13	8192	6
30	255.255.255.252	14	16384	2

Composition of an IPv4 address



IONOS



4.1 Case Study: Class C $\rightarrow$ 4 Subnets

Given:

Network: 200.200.200.0/24

Requirement: 4 subnets

- Borrowed bits = 2 ($2^2 = 4$)
- Remaining host bits = 6
- Block size = $2^6 = 64$

Subnet Ranges

Subnet	Subnet ID	First Host	Last Host	Broadcast
1	.0	.1	.62	.63
2	.64	.65	.126	.127
3	.128	.129	.190	.191
4	.192	.193	.254	.255

4.2 Case Study: Class C $\rightarrow$ 8 Subnets

- Borrowed bits = 3 ($2^3 = 8$)
- Remaining host bits = 5
- IPs per subnet = 32
- Valid hosts = 30

👉 Smaller subnet size, more subnets.

5. Key Rules, Formulas, and Exam Traps

Core Definitions

- **Subnet ID** → First address in subnet
- **Subnet DBA** → Last address in subnet
- **Valid Hosts** → Between Subnet ID and DBA

Formulas to Remember

- **Subnets** = 2^n
- **Hosts per subnet** = $2^h - 2$
- **Block Size** = $256 - \text{subnet mask octet}$

Address Usage Rules

Address Type	Source IP	Destination IP
Unicast	✓	✓
Network ID	✗	✗
DBA	✗	✓
LBA	✗	✓